



INFORMATION SHEET

Study title: A study on the consistency of Arterial Spin Labeling (ASL) imaging at different field strengths (1.5T vs. 3.0T)

You are invited to participate in the above project conducted by Dr. Shirley ZHANG, who is a staff member of the Department of Rehabilitation Sciences in The Hong Kong Polytechnic University. The project has been approved by the PolyU Institutional Review Board (PolyU IRB) (or its Delegate) (Reference Number: HSEARS20260317004).

Objectives:

- Primary objective: To evaluate the consistency of whole-brain CBF, arterial transit time (ATT), and arterial cerebral blood volume (aCBV) obtained from the multi-delay pCASL sequence at 1.5T and 3.0T field strengths.
- Secondary objective: To explore and quantify the range of differences in CBF values between the two field strengths and the possibility of establishing a correction model.
- Clinical significance: To provide supporting evidence for standardized ASL clinical examinations and research applications at 1.5T.

Background:

Arterial Spin Labeling (ASL) is a non-invasive magnetic resonance imaging (MRI) technique for assessing cerebral blood perfusion, applied broadly in neuroscience research and clinical diagnosis. In clinical practice, MR scanners of different field strengths (e.g., 1.5 Tesla and 3.0 Tesla) are widely used, and they differ in their imaging parameters. This necessitates a systematic evaluation of the consistency of ASL imaging results between the two field strengths.

MRI, which stands for Magnetic Resonance Imaging, is a type of imaging technology that uses magnetic fields and radio waves to create detailed images of the body's internal structures. Unlike X-rays and CT scans, MRI does not use ionizing radiation. MRI is used widely in medicine and is particularly useful for imaging soft tissues and organs like the brain, spinal cord, muscles, and heart. It's often used to diagnose a variety of medical conditions and to monitor how well the body is responding to treatment. It is a noninvasive and safe procedure for most people, although precautions are needed for individuals with certain types of implanted devices (like pacemakers) or who have metal in their bodies.

Participants: A total of 30 healthy volunteers will be recruited publicly from the community and hospital staff.

Inclusion and exclusion criteria:

Inclusion Criteria

1. Age 18–40 years;
2. No history of psychiatric or neurological disease;
3. No substantial intracranial parenchymal lesions detected on conventional MRI (e.g., cerebral infarction, brain tumor, white matter hyperintensities with a Fazekas grade ≥ 2 , or perivascular spaces [PVS] with a grade ≥ 3);
4. Willing to participate in this study and has signed the informed consent form.

Exclusion Criteria



1. Presence of contraindications to MRI (e.g., cardiac pacemaker, post intracranial aneurysm clipping, severe claustrophobia);
2. Pregnant or those planning pregnancy;
3. Failure to meet the pre-scan preparation requirements ;
4. Any other condition that, in the judgment of the investigator, makes the subject unsuitable for participation

Procedures:

This study is a prospective, self-controlled, cross-sectional study. Each subject serves as his or her own control and will undergo MRI scans twice, at 1.5T and 3.0T, respectively. The procedure is as follows:

1. Screening and Enrolment

1. Explain the purpose, procedures, and potential risks of the study to potential participants;
2. Obtain signatures for informed consent;
3. Collect demographic information (age, sex, and handedness), medical history, and medication history.

2. Pre-scan Preparation

1. To ensure the stability of the ASL signal, participants must strictly follow the requirements below:
2. Fasting: Participants should remain fasting for at least 2 hours since the last meal before scanning;
3. Dietary restrictions: On the day of scanning, no smoking, alcohol, coffee, strong tea, or other caffeine-containing foods;
4. Rest requirements: Avoid vigorous exercise, emotional agitation, and staying up late before scanning, and ensure adequate sleep;
5. Medication restrictions: Try to avoid taking medication for 3 days before scanning. If medication is unavoidable, the name and dosage of the medication taken should be documented.

3. Scanning Procedure

1. Each subject will complete MRI scans at 1.5T and 3.0T, with a short period (1-2 days) between the two scanning sessions;
2. Scan coverage: Ensure complete coverage of the whole brain, including the parietal lobes and cerebellum;
3. Image quality control: Image quality will be checked immediately after scanning to ensure that there are no obvious fold-over artifacts, motion artifacts, or other abnormalities affecting analysis. If the image quality is inadequate, rescanning may be considered as appropriate.

4. Data Collected

Basic information: Age, sex, handedness (left/right), and medication history.

Imaging data:

- 1.5T: Multi-delay pCASL, 3D-T1WI
- 3.0T: Multi-delay pCASL, 3D-T1WI
- Conventional T2WI/FLAIR (for lesion screening)

Potential risk or discomfort:



MRI uses a powerful magnet to take pictures of your body. Because the MRI machine exposes the body to a very strong magnetic force, you will have to follow certain safety precautions to make sure there are no metal objects in or on your body. Before you undergo the MRI scan, MR personnel will ask whether your body contains any metallic medical devices or equipment, including heart pacemakers, metal prostheses, implants or surgical clips. You also will be asked whether you have had any prior injuries from shrapnel or grinding metal, and if your eyes have been exposed to metal particles. If you have no metallic objects or particles in your body, you will need to remove all metal objects, including jewellery, watches, hair holders, or eyeglasses before entering the scanner room. You will need to empty your pockets of all materials, including keys, wallets, and magnetic cards such as ATM and credit cards. In addition, you will need to change into a patient gown. Finally, you may be asked to remove any eye shadow you may be wearing, because eye shadow sometimes contains metallic substances.

During the MRI examination, you will lie on a table that slides into a horizontal tube slightly wider than your body. You will be asked to lie still, but you will be able to hear and speak to the MRI personnel/research staff. Some individuals experience anxiety, panic, or a sensation of claustrophobia when lying in the MRI machine. If you think this may happen to you, please tell the MRI personnel before the scan. The scanner also makes loud noises during its operation. You will be provided with ear protection to reduce the noise level. If you feel uncomfortable for any reason before or during the procedure, please tell the MRI personnel. If for any reason during the procedure you want to stop, you may do so by pressing the squeeze ball.

Participant Responsibilities:

As a participant of a research study, your responsibilities include:

- Follow the instructions of the radiographer, MR personnel, and/or research staff.
- Inform the radiographer, MR personnel and/or research staff about any side effects, doctor visits, or hospitalizations that you may have.
- Inform the radiographer, MR personnel and/or research staff if you believe you might be pregnant.
- Inform the radiographer, MR personnel and/or research staff if you have some type of implanted electrical device (such as a cardiac pacemaker). You will not be allowed to participate in this study if you have one.
- Ask questions as you think of them.
- Inform the radiographer, MR personnel and/or research staff if you change your mind about staying in the study.
- Inform the radiographer, MR personnel and/or research staff if you are taking part in any other research project(s). This is to protect you from possible undesirable procedures or events such as extra blood drawing, extra x-rays, interaction(s) of research drugs, or other hazards.

Possible Discovery of Medical Findings:

When an MRI examination is performed, MR personnel may find something unexpected in the images. ***Since you are participating in a research project, the research team will not conduct a diagnostic review of the images and hence no written or verbal report of the results of the MRI examination, normal or otherwise, will be issued.***



The MRI Examination Does Not Substitute a Clinical Examination:

The MRI examination that you will receive is part of a research study. The researchers and The Hong Kong Polytechnic University do not bear responsibility for identifying or failing to identify abnormalities in the scan images. However, you may choose to be informed (or not to be informed) about incidental findings (if any) in the consent form. With this consent, the researchers may inform you when abnormalities are detected in your scan images. In no case will the MRI examination that you will undergo replace a visit to your own doctor. The images generated from the MRI examination will not be released to you for personal use.

Confidentiality:

The information you provide as part of the project is the research data. Any research data from which you can be identified is known as personal data. Personal data does not include data where the identity has been removed (anonymous data). We will minimize our use of personal data in the study as much as possible. The researcher and his team, supervisor will have access to personal data and research data for the purposes of the study. Responsible members of The Hong Kong Polytechnic University may be given access for monitoring and/or audit of the research. In accordance with our research data management policy, the data collected during this study will be retained for a period of 5 years from the date of the final publication based on the data.

All information related to you will remain confidential and will be identifiable by codes only known to the researchers. The information collected will be kept until five years after project completion/publication or public release of research results. The Hong Kong Polytechnic University takes reasonable precautions to prevent the loss, misappropriation, unauthorized access or destruction of the information you provide.

Participation and withdrawal:

Your decision to participate in this study is voluntary. You have every right to withdraw from the study before or during the measurement without penalty of any kind.

Contact:

If you have any questions, you may ask our helpers now or later, even after the study has started. You may contact **Dr Hui Zhang (tel. no.: 2766 4340; email: hui-shirley.zhang@polyu.edu.hk)** of PolyU under the following situations:

- if you have any other questions in relation to the study;
- if, under very rare conditions, you become injured as a result of your participation in the study; or
- if you want to get access to/or change your personal data before (the expiry date).

In the event you have any complaints about the conduct of this research study, you may contact **Ms. Vangie Chung (tel. no.: 27664329 email: vangie.chung@polyu.edu.hk)** or PolyU Institutional Review Board in writing (institutional.review.board@polyu.edu.hk) stating clearly the responsible person and department of this study as well as the Reference Number.



In case of a serious adverse event¹, please report to the Principal Investigator/Chief Investigator immediately and the Principal Investigator/Chief Investigator will be required to report it to the PolyU IRB within 48 hours upon the receipt of your report.

Thank you for your interest in participating in this study.

Dr. Hui Zhang, Shirley
Principal Investigator/Chief Investigator

¹ SAE is any adverse event that:

- Results in death
- Is life threatening, or places the participant at immediate risk of death from the event as it occurred
- Requires or prolongs hospitalization
- Causes persistent or significant disability or incapacity
- Results in congenital anomalies or birth defects
- Is another condition which investigators judge to represent significant hazards

(Reference: NIA Adverse Event and Serious Adverse Event Guidelines.
<https://www.nia.nih.gov/sites/default/files/2018-09/nia-ae-and-sae-guidelines-2018.pdf>)



研究題目：不同場強（1.5T vs 3.0T）下動脈自旋標記成像一致性研究

我們誠邀閣下參與由香港理工大學康復治療科學系的張慧博士進行的上述實驗項目。該實驗項目已獲得香港理工大學倫理審查委員會（或其代表）的批准（參考編號：HSEARS20260317004）。

目的：

- ❖ 主要目的：評估 1.5T 和 3.0T 場強下 Multi-delay pCASL 序列所測得全腦 CBF、動脈到達時間（ATT）及相對腦血容量（aCBV）參數的一致性。
- ❖ 次要目的：探索並量化兩種場強下 CBF 值的差異範圍，建立校正模型的可能性。
- ❖ 臨床意義：為在 1.5T 場強下開展標準化的 ASL 臨床檢查和科研應用提供依據。

背景：

動脈自旋標記（ASL）是一種無創腦血流灌注磁共振成像技術，在神經科學研究和臨床診斷中應用廣泛。由於臨床實踐中廣泛存在不同場強的磁共振設備（1.5T 和 3.0T），且兩者成像參數存在固有差異，有必要系統評估兩種場強下 ASL 成像結果的一致性。磁力共振造影是一種影像技術，它使用磁場和無線電波來創建人體內部結構的詳細影像。與 X 射線和電腦斷層掃描不同，磁力共振造影不使用電離輻射。磁力共振造影在醫學領域得到廣泛應用，尤其適用於成像軟組織和器官，如大腦、脊髓、肌肉和心臟。它通常用於診斷各種醫療狀況，並監測身體對治療的反應情況。對大多數人來說，磁力共振造影是一種非侵入性和安全的檢查方法，但對於某些類型的植入設備（如心臟起搏器）或體內有金屬的人需要採取預防措施。

參加者：

計劃通過公開招募的方式，從社區及醫院職工中招募 30 名健康志願者。

納入標準：

1. 年齡 18-40 歲;
2. 無精神疾病病史或神經系統疾病病史;
3. 常規 MRI 檢查未發現腦內實質性病變（例如：腦梗死、腦腫瘤，腦白質高信號 Fazekas 分級 ≥ 2 級，血管周圍間隙 PVS 分級 ≥ 3 級等）;
4. 自願參加本試驗並簽署知情同意書。

排除標準：

1. 存在 MRI 檢查禁忌症（如心臟起搏器、顱內動脈瘤夾閉術後、嚴重幽閉恐懼症等）;
2. 孕婦或備孕者;
3. 掃描前準備不符合要求者（見第 5 節）;
4. 經研究者判斷，存在其他不適合參加研究的情況。



實驗流程：本研究為**前瞻性、自身對照、橫向研究**。每位受試者自身作為對照，接受兩種場強（1.5T 和 3.0T）的 MRI 掃描，流程如下：

1 篩選與入組

1. 向潛在受試者解釋研究目的、流程及潛在風險；
2. 簽署書面知情同意書；
3. 收集人口學信息（年齡、性別、利手）、既往病史及用藥史。

2 掃描前準備

為確保 ASL 信號的穩定性，受試者需嚴格遵守以下準備要求：

1. **空腹：**掃描前保持空腹狀態（距離上次進食 ≥ 2 小時）；
2. **飲食限制：**掃描當天不抽煙、不喝酒、不喝咖啡及濃茶，且不攝入其他含咖啡因食品；
3. **作息要求：**掃描前避免劇烈運動和情緒激動，避免熬夜，保證充足睡眠；
4. **用藥限制：**掃描前 3 天盡量避免服用藥物，如確實需要服藥，需註明所服藥名及劑量。

3 掃描流程

1. 每位受試者將在同一天內先後完成 1.5T 和 3.0T 設備的掃描，兩次掃描之間允許短 1~2 天間隔。
2. **掃描範圍要求：**確保頂葉及小腦等全腦結構掃描完整。
3. **圖像質量控制：**掃描後即時檢查圖像質量，確保無明顯的卷褶偽影、運動偽影等影響分析的異常情況。若圖像不合格，視情況決定是否重掃。

4 數據採集內容

1. **基本信息：**年齡、性別、利手（左/右）、用藥史。
2. **影像數據：**
 - 1.5T: Multi-delay pCASL, 3D-T1WI
 - 3.0T: Multi-delay pCASL, 3D-T1WI
 - 常規 T2WI/FLAIR（用於病變篩查）

潛在風險或不適之處：

磁力共振造影掃描使用超強磁鐵來拍攝身體的影像。由於身體暴露在非常強的磁場下，因此您必須遵循安全預防措施，以確保身上或體內沒有金屬物體。在進行磁力共振造影掃描之前，研究人員將詢問您的身體是否含有任何金屬醫療設備，包括心臟起搏器、金屬假肢、金屬植入物或手術夾。您還將會被問及您是否曾被鋒利金屬劃傷，以及您的眼睛是否接觸過金屬顆粒。如果體內沒有金屬物體，則需要在進入掃描室之前移除所有身上金屬物體，包括珠寶、手錶、髮卡或眼鏡。您需要清空衣服口袋，包括鑰匙、錢包和磁卡（如信用卡）。此外，您將需要更換掃描服。最後，您可能會被要求去除您的眼影，因為眼影可能含有金屬物質。

在磁力共振造影掃描期間，您將躺在一張掃描床上。該掃描床會滑動至一個與身體平行的環形磁鐵中。您將躺在磁鐵中，並且能夠聽到研究人員說話的聲音。有些人在磁力共振造影掃描儀中會感到焦慮、恐慌或幽閉恐懼。如果您認為您可能發生此類情況，請在掃描前



告訴研究人員。掃描儀在操作過程中也會發出噪音。您可使用耳塞和耳罩，以降低噪音。如果您在掃描前或掃描中由於任何原因感到不舒服，請告訴研究人員。如果出於任何原因需要中斷掃描，請按壓緊急呼叫球。至今，沒有報告顯示磁力共振造影掃描檢查存在重大風險。然而，您應該知道，在這個領域可能存在其他未知的風險。

參與者職責：

作為研究的參與者，您的職責包括：

- 遵循放射技師、研究人員和/或其他工作人員的指示。
- 告知放射技師、研究人員和/或其他工作人員磁力共振造影檢查可能引發的任何副作用、就診或住院情況。
- 如果您認為您可能懷孕，請通知放射技師、研究人員和/或其他工作人員。
- 如果您有某種植入式電子設備（如心臟起搏器），請通知放射技師、研究人員和/或其他工作人員。如果您有此類設備，您將不能參加該項研究。
- 詢問您的任何疑惑。
- 如果您改變了參與該項研究的想法，請告知放射技師、研究人員和/或其他工作人員。
- 如果您正在參加任何其他研究項目，請告知放射技師、研究人員和/或其他工作人員。

意外發現：

進行磁力共振造影檢查時，研究人員可能會有意外的異常發現。**由於您參與的是學術研究項目（非醫療），研究團隊不會通過磁力共振影像進行醫學診斷，因此不會發出任何有關磁力共振檢查的書面或口頭報告。**

磁力共振造影檢查不能替代臨床檢查：

您將進行的磁力共振造影檢查是研究的一部分。研究人員和香港理工大學不承擔任何識別或未能識別掃描影像異常的責任。但是，您可以在同意書中選擇被告知（或不被告知）意外發現（如有）。經此同意，研究人員會通知您關於您磁力共振影像中任何發現（如有）。在任何情況下，您所接受的磁力共振造影檢查都不能代替醫生的檢查。磁力共振影像不會發給您供個人使用。

保密性：

閣下提供的資訊將作為該項研究的研究數據。可以通過這些數據識別閣下的數據被稱為個人數據。個人數據不包括已經去識別化的數據（匿名數據）。我們將盡可能減少在研究中使用個人數據的範圍。研究人員及其團隊、主管將可以訪問個人數據和研究數據，以開展研究目的。香港理工大學的負責成員可能會獲得訪問權限，用於監測和/或審核研究工作。根據我們的研究數據管理政策，本研究收集的數據將在最後出版的日期起計保留5年。

與閣下相關的所有信息將保密，僅研究人員知曉相應的編碼以識別閣下。收集的信息將保留到項目完成/發布研究結果的五年之後。香港理工大學將採取合理的預防措施，以防止閣下提供的信息丟失、被盜用、未經授權訪問或銷毀。

參與和退出：



閣下決定參與此研究是自願的。閣下有權在測量之前或期間自由退出研究，而不會受到任何懲罰。

聯繫方式：

如果閣下有任何問題，閣下可以現在或之後向研究員提問，甚至在研究開始後也可以。

以下情況下，閣下可以聯繫香港理工大學的張慧博士（電話：27664340，電子郵件：hui-shirley.zhang@polyu.edu.hk）：

- 如果閣下對研究有任何其他問題；
- 如果在極少數情況下，閣下因參與研究而受傷；
- 如果閣下希望在到期日之前獲取、更改閣下的個人數據。

如有對本研究的行為有任何投訴，閣下可以透過書面形式聯繫鍾靜妍女士（電話：27664329，電子郵件：vangie.chung@polyu.edu.hk）或香港理工大學機構審查委員會（電子郵件：institutional.review.board@polyu.edu.hk），清楚說明該研究的責任人和部門以及參考號碼。

在發生嚴重不良事件¹的情況下，請立即向首席研究員/主要研究員報告，首席研究員/主要研究員將需要在收到閣下的報告之後的 48 小時內向香港理工大學機構審查委員會報告。

感謝你對參與本研究的興趣。

張慧博士
首席研究員/主要研究員

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¹SAE（嚴重不良事件）是指任何不良事件，包括以下情況：

1. 導致死亡
2. 對參與者造成生命威脅，或使其立即處於死亡風險之中
3. 需要或延長住院時間
4. 導致持續或重大的殘疾或無能力
5. 導致先天畸形或出生缺陷
6. 被調查人員判斷具有重大危害的其他情況（參考：NIA 不良事件和嚴重不良事件指南。
<https://www.nia.nih.gov/sites/default/files/2018-09/nia-ae-and-sae-guidelines-2018.pdf>）

CONSENT TO PARTICIPATE IN RESEARCH

A study on the consistency of Arterial Spin Labeling (ASL) imaging at different field strengths (1.5T vs. 3.0T)

I _____ hereby consent to participate in the captioned research conducted by Dr. Hui Zhang. I understand that information obtained from this research may be used in future research and published. However, my right to privacy will be retained, i.e. my personal details will not be revealed. The procedure as set out in the attached information sheet has been fully explained. I understand the benefits and risks involved. My participation in the project is voluntary. I acknowledge that I have the right to question any part of the procedure and can withdraw at any time without penalty of any kind.

I affirm that I do not have a cardiac/heart pacemaker, brain aneurysm clips, any implanted metallic or electrical device, or any internal metal foreign object(s).

If there is any incidental finding, I:

- ☐ wish to be informed by the principal investigator (PI)
- ☐ do NOT wish to be informed.

I can contact one of the investigators, Dr. Hui Zhang at 27664340, about this study.

If I have complaints related to the investigator(s), I can contact Ms. Vangie Chung (tel. no.: 27664329 email: Vangie.Chung@polyu.edu.hk). I know I will be given a signed copy of this consent form.

By signing this form, I willingly agree to participate in the research it describes.

Please note that the MRI examination does not substitute a clinical examination.

Name of participant: _____

Signature of participant: _____

Name of researcher: _____

Signature of researcher: _____

Date: _____

參與研究同意書

不同場強（1.5T vs 3.0T）下動脈自旋標記成像一致性研究

本人_____同意參與由張慧博士開展的上述研究。本人知悉此研究所得的資料可能被用作日後的研究及發表，但本人的私隱權利將得以保留，即本人的個人資料不會被公開。研究人員已向本人清楚解釋列在所附資料卡上的研究程序，本人明瞭當中涉及的利益及風險；本人自願參與研究項目。本人知悉本人有權就程序的任何部分提出疑問，並有權隨時退出而不受任何懲處。

本人確認我沒有心臟起搏器，腦動脈瘤鉗，任何植入的金屬或電子設備，或任何體內金屬異物。

如有偶然發現，本人：

- ☐ 希望得到主要研究者的通知；
- ☐ 不希望得到主要研究者的通知。

關於這項研究，本人可以聯繫其中一位研究人員，張慧博士，電話號碼：27664340，詢問有關這項研究的問題。如果本人對研究人員有投訴，本人可以聯繫鍾靜妍女士（電話：27664329，電子郵件：Vangie.Chung@polyu.edu.hk）。本人知道本人會收到這份同意書的簽署副本。

在此表格上簽字，表示本人願意參與其中所描述的研究。

請注意，磁力共振檢查不能代替臨床檢查。

參與者姓名：_____

參與者簽署：_____

研究人員姓名：_____

研究人員簽署：_____

日期：_____